MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

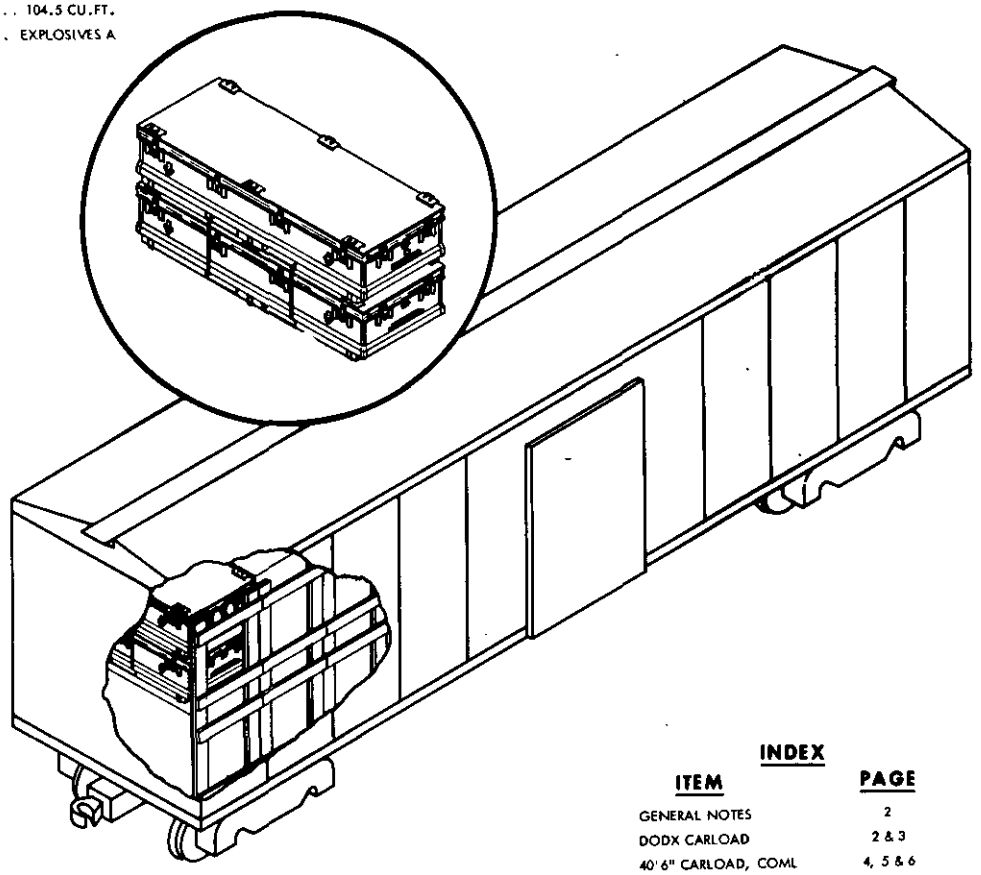
MIL-STD-1325-168
(NAVY)

18 JANUARY 1974

BOMB, CLUSTER, MK 20 AND MODS (ROCKEYE II) IN CONTAINER CNU-208/E FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/250
WEIGHT (APPROX.) . . . . . 3220 LBS
DIMENSIONS . . . . . 102 L x 36.5 W x 48.5 H (47.5 STACKING HEIGHT)
CUBE . . . . . 104.5 CU.FT.
DOT HAZARD CLASSIFICATION . . EXPLOSIVES A



INDEX

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DODX CARLOAD	2 & 3
40'6" CARLOAD, COML	4, 5 & 6
50'-6" CARLOAD, COML	8, 9 & 10

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M. S. Gray NWHL — 1/15/74
SIGNATURE — TECHNICAL DIRECTION AGENT (TDA) — DATE

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APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck — 12/18/73
SIGNATURE — SUPERVISOR, MILITARY & INTERMODEL SERVICES — DATE

ORIGINATOR Charles McDade 1/15/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

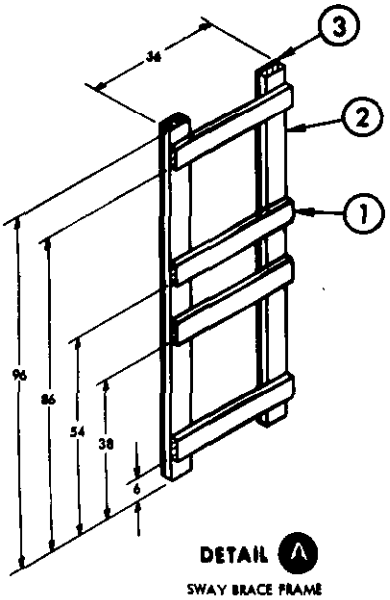
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE *THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS* MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. *THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE* "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF BOMB, CLUSTER, MK 20 AND MODS, (ROCKEYE II) IN CONTAINER CNU-208/E PALLETIZED IN ACCORDANCE WITH WR 54/250.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORKLIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. *APPLICABLE MATERIAL SPECIFICATIONS:*
   DUNNAGE LUMBER - FED. SPEC MM-L-751
   NAILS - FED. SPEC FF-N-105
   STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
   SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

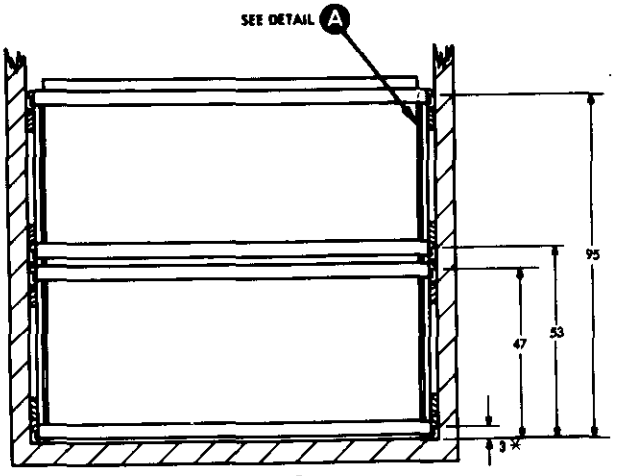
50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 28 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.



DETAIL A

SWAY BRACE FRAME
28 REQUIRED



SECTION B-B

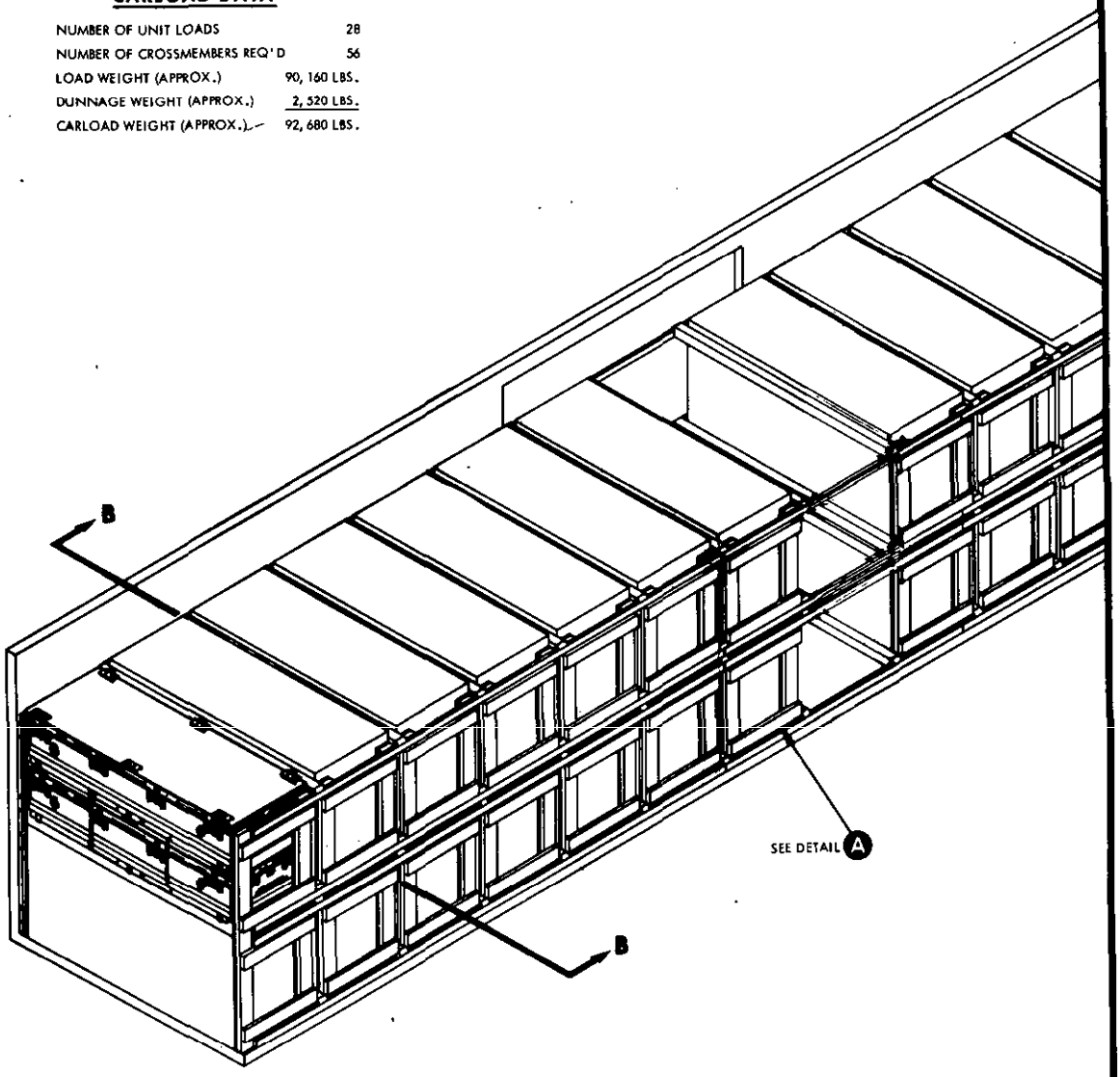
SHOWING LOCATION OF WALL MEMBERS AND DOORWAY MEMBERS USED FOR CROSSMEMBERS

*5" IN DOORWAY AREA

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CARLOAD DATA

NUMBER OF UNIT LOADS	28
NUMBER OF CROSSMEMBERS REQ'D	56
LOAD WEIGHT (APPROX.)	90, 160 LBS.
DUNNAGE WEIGHT (APPROX.)	2, 520 LBS.
CARLOAD WEIGHT (APPROX.)	92, 680 LBS.



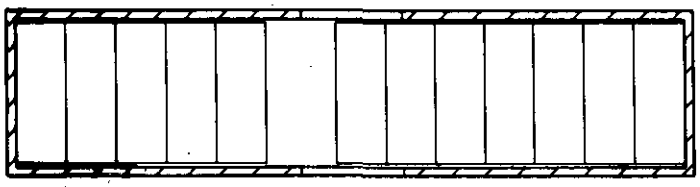
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
3	SWAY BRACE VERTICAL	1 x 6 x 96	56	2	1 PER FOOT	6d
2	SWAY BRACE VERTICAL	2 x 6 x 96	56	1	3 PER JOINT	10d
1	SWAY BRACE HORIZONTAL	2 x 6 x 36	112	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 24 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT. WIDE.
4. WHEN BOXCARS OTHER THAN 9 FT. 2 IN. WIDE ARE USED, PIECES 2 OR 3 SHOULD BE INCREASED OR DECREASED AS NECESSARY FOR A TIGHT LOAD ACROSS THE CAR.



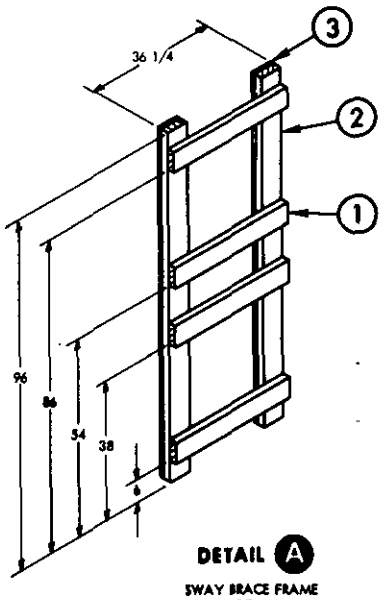
LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
12	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	10	11	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 x 3 x 108	4	DOOR POST	2 PER FOOT	20d
10	HOLD DOWN CLEAT	2 x 6 x 24	4	9	5	10d
9	GATE HOLD DOWN	2 x 6 -12 FT	2	CAR WALL	5 EACH END	10d
8	STRUT	4 x 4 x WEDGE FIT *	20	5	2 EACH END	16d
7	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	5	3 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	10	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 x 6 x 100	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	10	5	3 PER JOINT	10d
3	SWAY BRACE VERTICAL	1 x 6 x 96	48	2	1 PER FOOT	6d
2	SWAY BRACE VERTICAL	2 x 6 x 96	48	1	3 PER JOINT	10d
1	SWAY BRACE HORIZONTAL	2 x 6 x 36 1/4	96	SEE 2	-	-

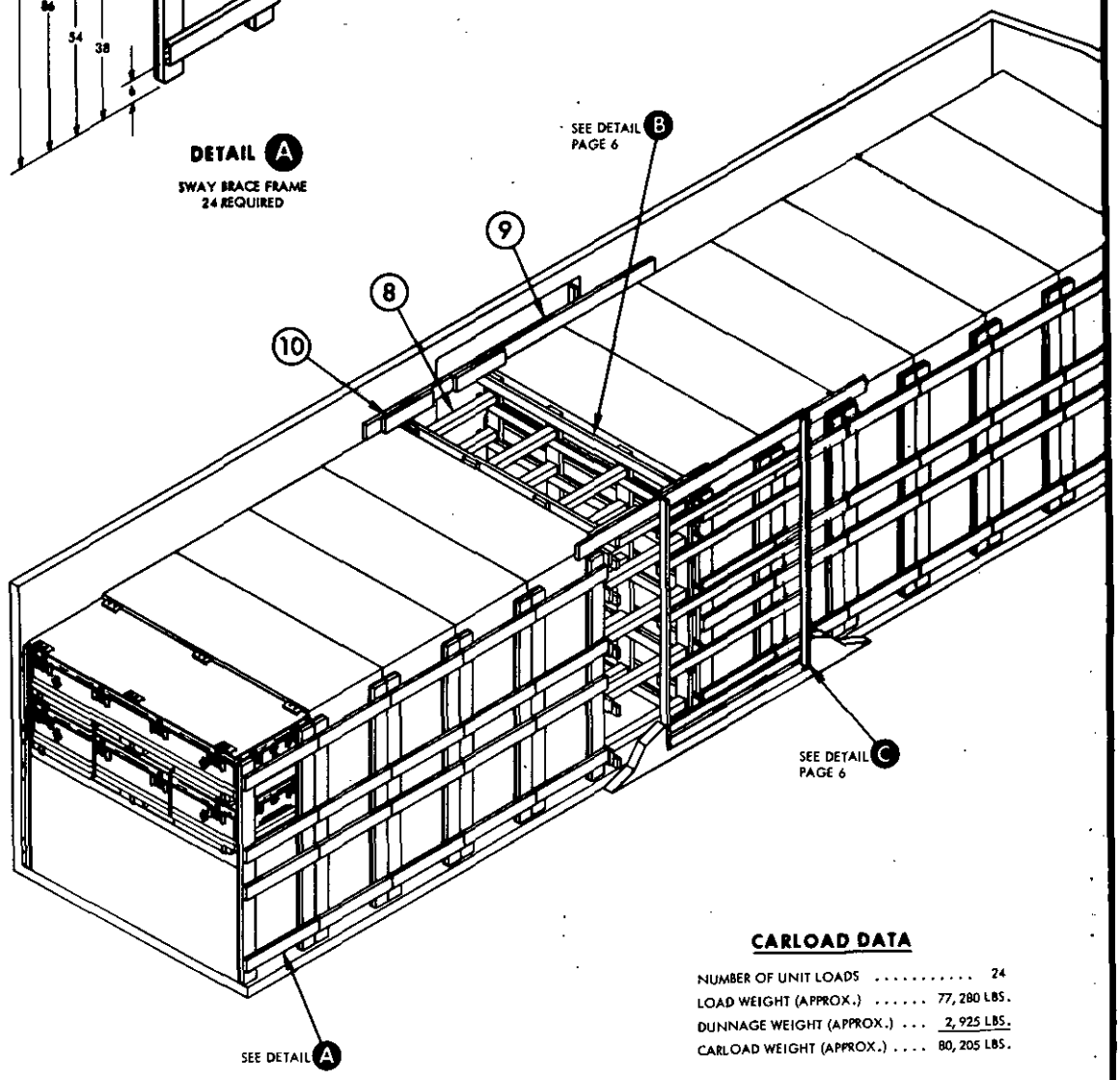
LIST OF MATERIALS AND NAILING DATA

* 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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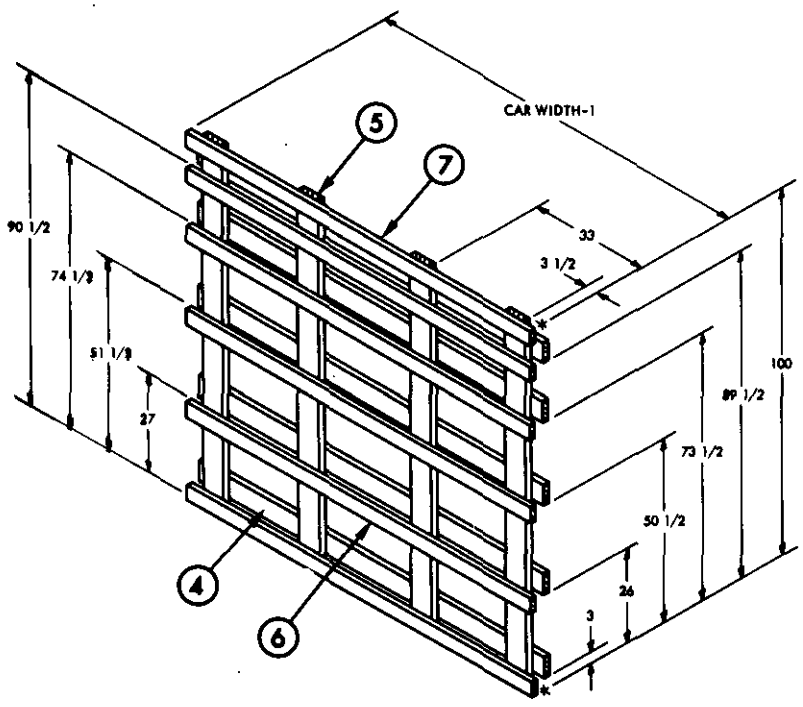
DETAIL A
SWAY BRACE FRAME
24 REQUIRED



CARLOAD DATA

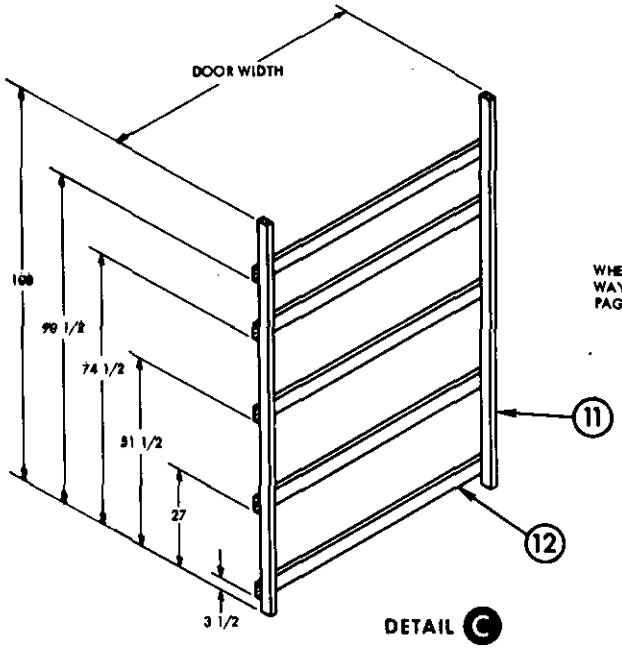
NUMBER OF UNIT LOADS	24
LOAD WEIGHT (APPROX.)	77,280 LBS.
DUNNAGE WEIGHT (APPROX.)	2,925 LBS.
CARLOAD WEIGHT (APPROX.)	80,205 LBS.

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DETAIL B

CENTER GATE
2 REQUIRED



WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

DETAIL C

DOORWAY PROTECTION
2 REQUIRED

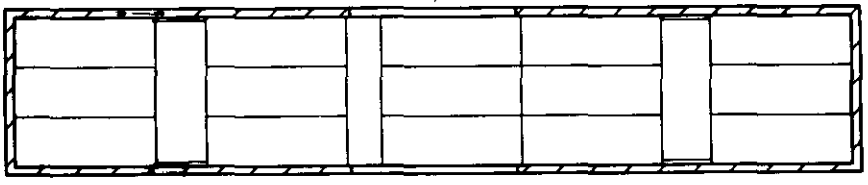
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF EITHER 34 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING, OR, IF THE INSIDE WIDTH OF THE RAILCAR DOES NOT PERMIT LOADING THREE ROWS ACROSS THE CAR, 32 UNIT LOADS WHICH MUST BE LOADED IN ACCORDANCE WITH THE ALTERNATE LOADING PLAN SHOWN ON PAGE 10. WHEN USING THE ALTERNATE LOADING PLAN 32 SWAY BRACE FRAMES, DETAIL A, PAGE 9 ARE REQUIRED AND THE CENTER GATE SHOWN ON PAGE 6 SHOULD BE USED IN PLACE OF DETAIL B, PAGE 10.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 8 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 8 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND DOORWAY PROTECTION, PIECES 10 AND 11, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. WHEN BOXCARS OTHER THAN 9 FT 2 IN. WIDE ARE USED, PIECES 2 OR 3 SHOULD BE INCREASED OR DECREASED AS NECESSARY FOR A TIGHT LOAD ACROSS THE CAR.

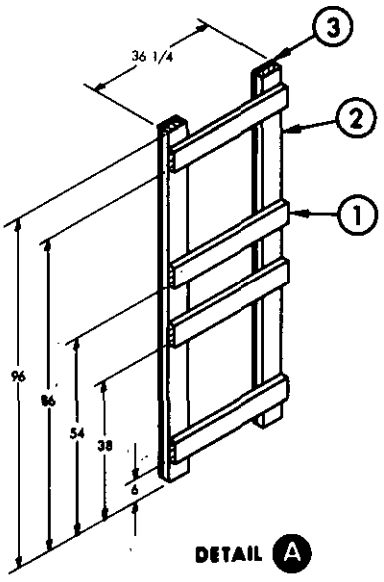


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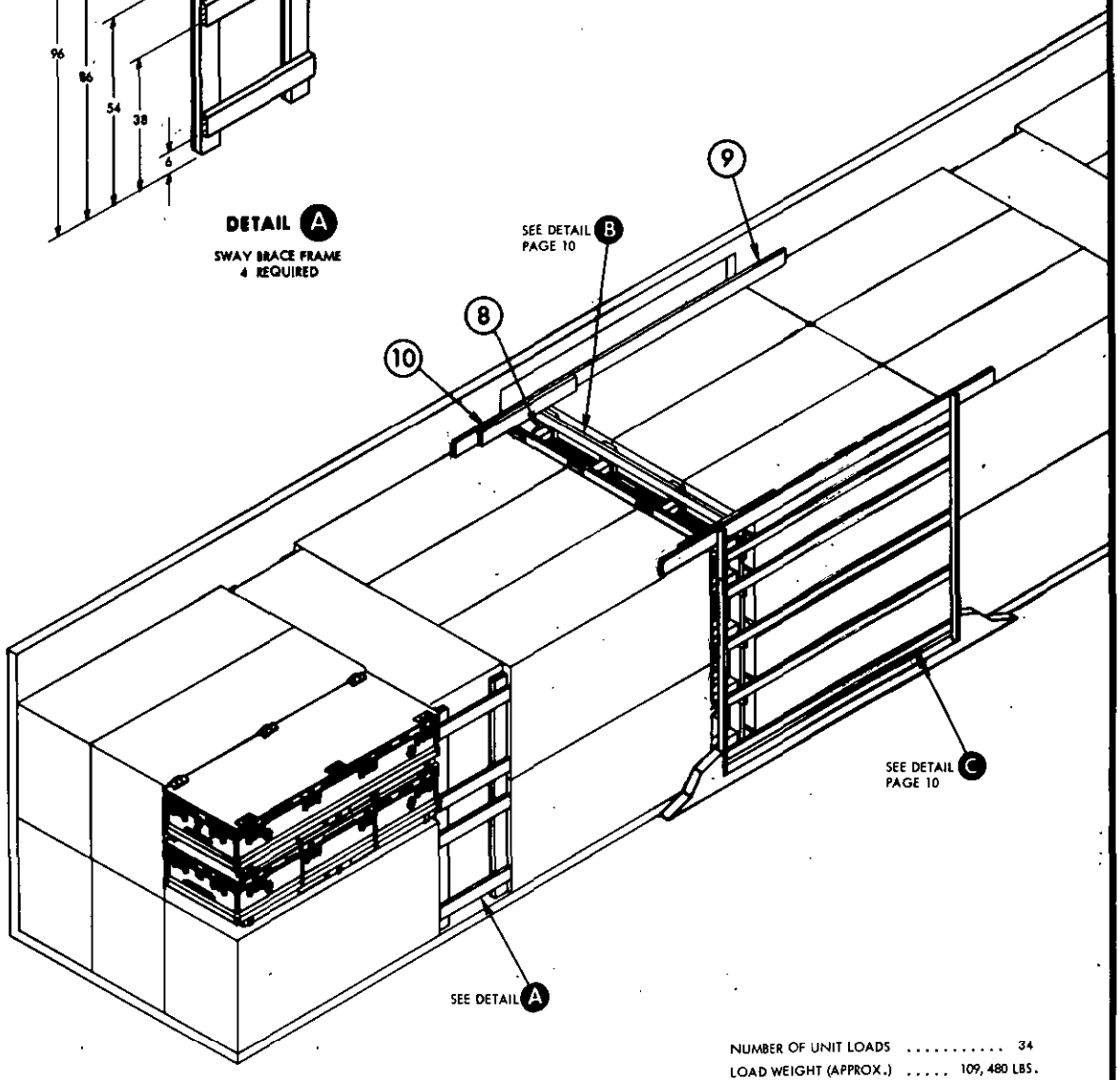
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
	LIST OF MATERIALS AND NAILING DATA					
12	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	10	10	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 x 3 x 108	4	DOOR POST	2 PER FOOT	20d
10	HOLD DOWN CLEAT	2 x 6 x 48	2	8	2 PER FOOT	10d
9	GATE HOLD DOWN	2 x 6-14 FT	2	CAR WALL	5 EACH END	10d
8	STRUT	4 x 4 x WEDGE FIT *	20	5	2 EACH END	16d
7	TOP HORIZONTAL	2 x 4 x CAR WIDTH - 1	2	5	3 PER JOINT	10d
6	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	10	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 x 6 x 100	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	10	5	3 PER JOINT	10d
3	SWAY BRACE VERTICAL	1 x 6 x 96	8	2	1 PER FOOT	6d
2	SWAY BRACE VERTICAL	2 x 6 x 96	8	1	3 PER JOINT	10d
1	SWAY BRACE HORIZONTAL	2 x 6 x 36	16	SEE 2	-	-

* 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAYBE SUBSTITUTED IN PLACE OF 4 x 4'S.

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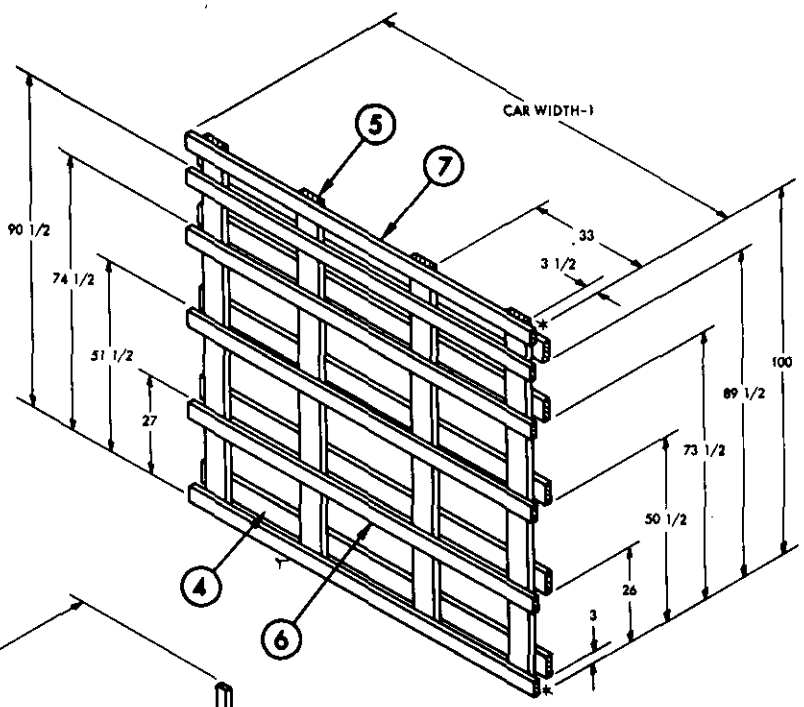


DETAIL A
SWAY BRACE FRAME
4 REQUIRED

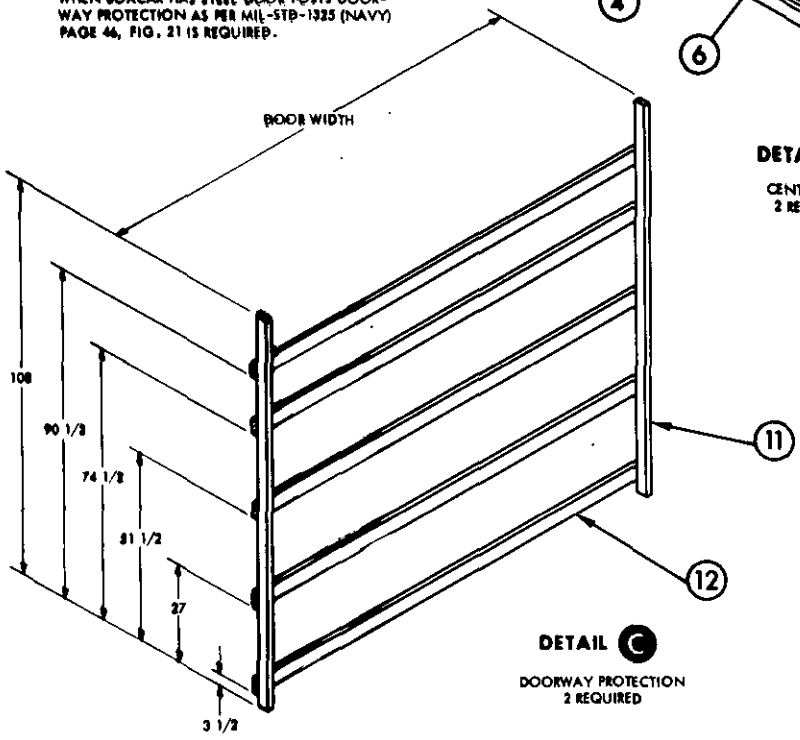


NUMBER OF UNIT LOADS	34
LOAD WEIGHT (APPROX.)	109, 480 LBS.
DUNNAGE WEIGHT (APPROX.)	1, 271 LBS.
CARLOAD WEIGHT (APPROX.)	110, 751 LBS.

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WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

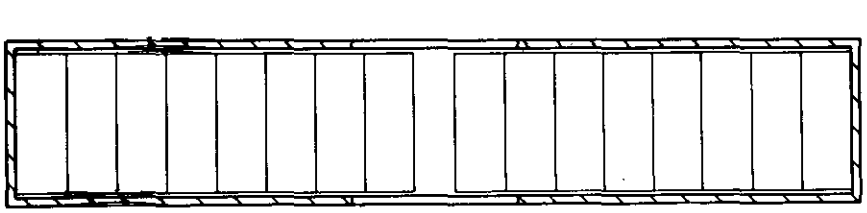


DETAIL B

CENTER GATE
2 REQUIRED

DETAIL C

DOORWAY PROTECTION
2 REQUIRED



LOADING PLAN

REVIEW ACTIVITY
NAVY - OS, AS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 8140-N219)

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